Supplementary Material S3: Theorem Cards for the Six Formal Results

Recursive Training on Model-Generated Data: A Critical Integrative Review

of Failure Regimes, Adversarial Interactions and Governance

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Contents

S3. Theorem cards for formal results

For each of the six formal results summarised in Section 2.1 and Table 2 of the main text, this supplement lists: the source and version consulted, the model class, data regime, loss function, asymptotic variable, collapse criterion and main assumptions, and the principal "does not imply" caveat stated in the main text.

Result 1: full replacement leads to generation-wise error growth.

Sources Dohmatob et al. (2024a), Theorem 1 of arXiv:2402.07043 (original result); Gerstgrasser et al. COLM 2024 (arXiv:2404.01413, Eq. (4)) restates it to contrast with the accumulation bound.

Model class Ordinary least squares (Gaussian linear regression).

Regime Replace (no real data after generation 0); equal sample size N at each generation.

Loss / asymptotic variable Quadratic loss; asymptotic in generations g.

Design Fixed initial real dataset $X \in \mathbb{R}^{N\times d}$; same design matrix used for synthetic target generation at every generation.

Assumptions $N > d + 1$; isotropic design $\Sigma = I_d$; independent noise σ^2 per label.

Main result $E_{\text{test}}^{\text{Replace}}(\hat{w}_g) = E_0 \cdot g$, where $E_0 = \sigma^2 d/(N - d - 1)$.

Does not imply Error growth when real data are retained (Result 2); or that neural scaling laws are invalid for frontier LLMs trained on mixed corpora.

Result 2: accumulation bounds the error.

Sources Gerstgrasser et al. COLM 2024 (arXiv:2404.01413, Theorem 1); empirical support across architectures and datasets in Kazdan et al. ICML 2025 (arXiv:2410.16713).

Model class As Result 1.

Regime Accumulate (original real data retained and accumulated alongside each new generation of synthetic data).

Design Fixed initial real dataset (same X accumulated at every generation).

Main result $E_g = E_0 \sum_{i=1}^{g+1} i^{-2} < E_0\pi^2/6 \approx 1.645\, E_0$.

Numerical check Supplementary Material S4 confirms the equality form holds under the fixed design and shows the bound is not attained with fresh inputs.

Does not imply Bounded error under temperature-biased sampling (Result 6); or that the synthetic fraction is bounded (it can approach 1 under accumulation while the bound still holds).

Result 3: a fixed synthetic fraction produces an error floor.

Source Dohmatob et al. ICLR 2025 (arXiv:2405.05846), Theorem 3.2.

Model class High-dimensional linear regression (single training run, non-recursive); covariate distribution non-isotropic.

Regime Fixed-mix: a fraction $p \in (0, 1)$ of training data is synthetic in a single run.

Loss / asymptotic variable Quadratic loss; asymptotic in training set size N (not in generations).

Main result Non-vanishing excess error floor $\Delta(p) > 0$ determined by p and the gap between real and synthetic distributions; floor $\to 0$ as $p \to 0$.

Does not imply That this floor applies in multi-generation recursive training (different model from Results 1–2); or that neural scaling laws are invalid for frontier LLMs.

Result 4: without real data, the generative distribution degenerates.

Sources Shumailov et al. Nature 2024 (original qualitative result, Gaussian mixture and language model experiments); Seddik et al. COLM 2024 (arXiv:2404.01413-bis, Theorem 1), statistical LM: model collapse cannot be avoided under fully synthetic training; probability mass concentrates and tails disappear; Suresh et al. 2025 (arXiv:2412.17646, Theorem 1): for discrete distributions, the expected number of symbols surviving after k iterations is $\sum_i \exp(-np_i g_k)$, where p_i is the original frequency; survival time $\approx np_i$ generations.

Model class Language / discrete / Gaussian models; fully synthetic regime.

Regime Replace (fully synthetic; no real data after generation 0).

Loss Maximum likelihood / cross-entropy.

Main result Tails of the original distribution disappear; symbols survive for approximately $n \cdot p_i$ generations (Theorem 1 of Suresh et al.); full degeneration to a limited vocabulary or singular distribution under repeated replacement (Seddik et al., Shumailov et al.).

Does not imply That collapse is unavoidable with any real data (see Results 2–3); or a specific rate for LLMs trained on web-scale corpora.

Result 5: a random-walk view of parameter drift.

Source Xu et al. arXiv:2505.13947v1 (2025). Theorems used: Theorem 1 (divergence under constant n, Gaussian variance estimation); Theorem 3 (sufficiency of $c_t = t^{1+s}$ for unbiased estimators with $\kappa \geq \gamma/2$); Theorem 5 ($P(T) < 1/2$ for any schedule under asymptotic normality).

Model class Generic exponential family under uniform tail-bound Assumption 1; unbiasedness required for Theorem 3 (Assumption 2).

Regime Fully synthetic; varying sample schedule $n_t = c_t \cdot n$.

Asymptotic variable Number of recursive training steps T; limit taken jointly as $n \to \infty$ and $T \to \infty$.

Main results
- *Theorem 1* (Gaussian variance, constant n): $E[(\hat{\sigma}_T^2 - \sigma^2)^2] \to \infty$ as $T \to \infty$; diversity vanishes with probability 1.
- *Theorem 3* (general unbiased, $\kappa \geq \gamma/2$): schedule $c_t = t^{1+s}$ ($s > 0$) satisfies $\lim_{n\to\infty} \lim_{T\to\infty} P(\|\hat{\theta}_T - \theta^*\|_2 \geq \delta) = 0$ for any $\delta > 0$.
- *Theorem 5* ($P(T)$ bound): under asymptotic normality, the probability that the T-th synthetic-data estimate outperforms the real-data estimate is $< \frac{1}{2}$ for any T and any sample schedule.

Does not imply Negative expected return on synthetic training (Theorem 5 bounds probability, not expectation); coexistence of bounded error and useful synthetic training illustrated in S4.2; the $c_t = t^{1+s}$ schedule is sufficient (Theorem 3), not necessary.

Result 6: temperature-filtered sampling under accumulation.

Source Marchi et al. IEEE CDC 2024 (DOI: 10.1109/CDC56724.2024.10886816), Theorems 1–3.

Model class Discrete generative model with finite output set; closed-loop maximum likelihood or cross-entropy; exact-tracking limit ($\delta = 0$ per-step approximation error).

Regime Accumulate; original real data retained.

Asymptotic variable Number of generated samples (asymptotic in size of synthetic batches per generation, not in number of generations).

Main results
- $T_s = 1$ (neutral sampling): dynamics converge to a finite drift; the accumulation bound of Result 2 holds.
- $T_s \neq 1$ (temperature-biased sampling): distribution converges to the uniform distribution on a subset of the vocabulary, or to the simplex boundary, depending on T_s and the initial distribution.

Does not imply That accumulation fails generally; only that it fails when sampling is temperature-biased. The closed-loop approximation ($\delta = 0$) is an idealisation not achieved in practice.